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ZEEL V0.2.1

Zeek Cheat Sheet

A quick-reference guide for Zeek (formerly Bro): command-line usage, core logs, connection states, history letters, and key log fields for network detection and response.



Useful Zeek Commands

TASK	COMMAND
Analyze a PCAP	<code>zeek -r capture.pcap</code>
Live capture	<code>zeek -i eth0</code>
Ignore checksums	<code>zeek -C -r file.pcap</code>
Set the local network	<code>zeek -r file.pcap "Site::local_nets += { 10.0.0.0/8 }"</code>
Extract all files from PCAPs	<code>zeek -C -r *.pcap /usr/local/share/zeek/policy/frameworks/files/extract-all-files.zeek</code>
Extract transferred files	<code>zeek -r file.pcap -e "event file_new(f: fa_file) { files::add_analyzer(f, Files::ANALYZER_EXTRACT); }"</code>
Load policy scripts	<code>zeek -r file.pcap policy/protocols/conn/known-hosts</code>
View the conn.log	<code>cat conn.log</code>
Read compressed logs	<code>zcat conn.log.gz less</code>
Show connections as a table	<code>zeek-cut ts uid id.orig_h id.orig_p id.resp_h id.resp_p proto conn_state < conn.log</code>
List unique originators	<code>zeek-cut id.orig_h < conn.log sort -u</code>
Count connections	<code>wc -l conn.log</code>
Filter DNS queries	<code>zeek-cut ts query qtype_name rcode_name < dns.log head</code>
HTTP status failures	<code>zeek-cut ts method host uri status_code < http.log awk '\$5 >= 400'</code>
Find notices	<code>grep Notice notice.log head</code>
Deploy via zeekctl	<code>zeekctl deploy</code>

Command-Line Usage

COMMAND	PURPOSE
<code>grep -H -P ""#fields" *.log</code>	List fields in all log files.
<code>zeek -r capture.pcap</code>	Analyze a pcap file offline (writes logs).
<code>zeek -i eth0</code>	Capture and analyze live traffic on an interface.
<code>zeek -C -r file.pcap</code>	Ignore invalid IP/TCP checksums.
<code>zeek -e "event ..."</code>	Run a Zeek event expression directly.
<code>zeek -s signatures.sig</code>	Load a signatures file.
<code>zeek -l dir</code>	Write logs to a local directory.
<code>zeek -N</code>	List loaded modules / analyzers.
<code>zeek-cut field1 field2 ... < log</code>	Extract specific fields from a log file.
<code>zeek-cut -d</code>	Convert time values into human readable format
<code>zeek-cut -n</code>	Print all fields "except" those specified (negate selection).
<code>zeek-cut -u</code>	Similar to -d, but print timestamps in UTC instead of local.

Core Logs Reference

LOG	WHAT IT LOGS
<code>conn.log</code>	IP, TCP, UDP, and ICMP connections.
<code>dns.log</code>	DNS queries and responses.
<code>http.log</code>	HTTP requests and replies.
<code>ssl.log</code>	TLS/SSL handshakes, ciphers, and cert.
<code>x509.log</code>	X.509 certificate details and fingerprints.
<code>files.log</code>	File analysis results (MIME, hashes).
<code>notice.log</code>	Interesting events and activity (alerts).
<code>weird.log</code>	Unexpected or anomalous activity.
<code>ssh.log</code>	SSH handshakes and authentication
<code>dhcp.log</code>	DHCP lease activity.
<code>ftp.log</code>	FTP commands and replies.
<code>smtp.log</code>	SMTP email transactions.
<code>ntlm.log</code>	NTLM authentication.
<code>kerberos.log</code>	Kerberos authentication (AS/TGS).
<code>rdp.log</code>	Remote Desktop Protocol activity.
<code>pe.log</code>	Portable executable metadata.
<code>syslog.log</code>	Syslog messages.
<code>dce_rpc.log</code>	DCE/RPC calls.
<code>radius.log</code>	RADIUS authentication attempts.
<code>sip.log</code>	SIP calls.
<code>snmp.log</code>	SNMP messages.
<code>socks.log</code>	SOCKS proxy requests.
<code>mysql.log</code>	MySQL commands.
<code>irc.log</code>	IRC communication.
<code>ldap.log</code>	LDAP transactions.
<code>quic.log</code>	QUIC connection updates.
<code>tunnel.log</code>	Encapsulating tunnels (GRE, IP-in-IP, etc.).
<code>websocket.log</code>	WebSocket handshakes.
<code>analyzer.log</code>	Protocol analysis violations.

http.log

FIELD	DESCRIPTION
<code>ts</code>	Time of the request
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>trans_depth</code>	Pipelined request depth
<code>method</code>	HTTP verb
<code>host</code>	Host header
<code>uri</code>	Request URI
<code>referrer</code>	Referer header
<code>version</code>	Reply HTTP version
<code>user_agent</code>	User-Agent header
<code>origin</code>	Origin header
<code>request_body_len</code>	Uncompressed client body size
<code>response_body_len</code>	Uncompressed server body size
<code>status_code</code>	Server status code
<code>status_msg</code>	Server status message
<code>info_code</code>	Last 1xx informational code
<code>info_msg</code>	Last 1xx informational message
<code>tags</code>	Discovered attributes
<code>username</code>	Client username
<code>password</code>	Captured password
<code>proxied</code>	Proxy-related headers
<code>orig_fuids</code>	Client file UIDs
<code>orig_filenames</code>	Client filenames
<code>orig_mime_types</code>	Client MIME types
<code>resp_fuids</code>	Server file UIDs
<code>resp_filenames</code>	Server filenames
<code>resp_mime_types</code>	Server MIME types

sip.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>trans_depth</code>	Pipelined transaction depth
<code>method</code>	SIP verb
<code>uri</code>	Request URI
<code>date</code>	Date header
<code>request_from</code>	Request From header
<code>request_to</code>	Request To header
<code>response_from</code>	Response From header
<code>response_to</code>	Response To header
<code>reply_to</code>	Reply-To header
<code>call_id</code>	Call-ID
<code>seq</code>	CSeq
<code>subject</code>	Subject header
<code>request_path</code>	Request Via path
<code>response_path</code>	Response Via path
<code>user_agent</code>	User-Agent
<code>status_code</code>	Status code
<code>status_msg</code>	Status message
<code>warning</code>	Warning header
<code>request_body_len</code>	Request Content-Length
<code>response_body_len</code>	Response Content-Length
<code>content_type</code>	Server Content-Type

dns.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>proto</code>	Transport protocol
<code>trans_id</code>	DNS transaction ID
<code>rtt</code>	Query-to-response RTT
<code>query</code>	Queried domain name
<code>qclass</code>	Query class value
<code>qclass_name</code>	Query class name
<code>qtype</code>	Query type value
<code>qtype_name</code>	Query type name
<code>rcode</code>	Response code value
<code>rcode_name</code>	Response code name
<code>AA</code>	Authoritative answer
<code>TC</code>	Truncated
<code>RD</code>	Recursion desired
<code>RA</code>	Recursion available
<code>Z</code>	Reserved Z field
<code>answers</code>	Answer resource records
<code>TTLs</code>	Caching intervals of answers
<code>rejected</code>	Query rejected by server
<code>opcode</code>	DNS opcode value
<code>opcode_name</code>	DNS opcode name

files.log

FIELD	DESCRIPTION
<code>ts</code>	Time file was first seen
<code>fuid</code>	File UID
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>source</code>	Analyzer that produced the file
<code>depth</code>	Encapsulation depth
<code>analyzers</code>	File analyzers applied
<code>mime_type</code>	Detected MIME type
<code>filename</code>	Filename from the source
<code>local_orig</code>	Originator is local
<code>is_orig</code>	File sent by originator
<code>seen_bytes</code>	Bytes seen by analysis
<code>total_bytes</code>	Expected full file size
<code>missing_bytes</code>	Bytes missed (packet loss)
<code>overflow_bytes</code>	Bytes not delivered to stream analyzers
<code>timeout</code>	Analysis timed out
<code>parent_fuid</code>	Parent container file UID
<code>md5</code>	MD5 digest
<code>sha1</code>	SHA1 digest
<code>sha256</code>	SHA256 digest
<code>extracted</code>	Extracted local filename
<code>extracted_cutoff</code>	Extraction was truncated
<code>extracted_size</code>	Bytes written to disk

smtp.log

FIELD	DESCRIPTION
<code>ts</code>	Time message first seen
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>trans_depth</code>	Message transaction depth
<code>helo</code>	HELO/EHLO contents
<code>mailfrom</code>	MAIL FROM address
<code>rcptto</code>	RCPT TO addresses
<code>date</code>	Date header
<code>from</code>	From header
<code>to</code>	To header
<code>cc</code>	CC header
<code>reply_to</code>	Reply-To header
<code>msg_id</code>	Message ID
<code>in_reply_to</code>	In-Reply-to
<code>subject</code>	Subject
<code>x_originating_ip</code>	X-Originating-IP
<code>first_received</code>	First Received header
<code>second_received</code>	Second Received header
<code>last_reply</code>	Last server reply
<code>path</code>	Transmission path from headers
<code>user_agent</code>	User-Agent
<code>tls</code>	Connection switched to TLS
<code>fuids</code>	Attached file UIDs

conn.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>proto</code>	Transport protocol
<code>service</code>	Detected application protocol(s)
<code>duration</code>	Connection duration
<code>orig_bytes</code>	Payload bytes from originator
<code>resp_bytes</code>	Payload bytes from responder
<code>conn_state</code>	Connection state
<code>local_orig</code>	Originator is local
<code>local_resp</code>	Responder is local
<code>missed_bytes</code>	Bytes lost to content gaps
<code>history</code>	Connection history flags
<code>orig_pkts</code>	Packets from originator
<code>orig_ip_bytes</code>	IP-layer bytes from originator
<code>resp_pkts</code>	Packets from responder
<code>resp_ip_bytes</code>	IP-layer bytes from responder
<code>tunnel_parents</code>	Parent tunnel connection UIDs
<code>ip_proto</code>	IP header protocol number

rdp.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>cookie</code>	Client cookie (often truncated username)
<code>result</code>	Negotiation/GCC status
<code>security_protocol</code>	Protocol chosen by server
<code>client_channels</code>	Channels requested by client
<code>keyboard_layout</code>	Client keyboard language
<code>client_build</code>	RDP client version
<code>client_name</code>	Client machine name
<code>client_disg_product_id</code>	Client product ID
<code>desktop_width</code>	Client desktop width
<code>desktop_height</code>	Client desktop height
<code>requested_color_depth</code>	Requested color depth
<code>cert_type</code>	Native RDP cert type
<code>cert_count</code>	Native RDP cert count
<code>cert_permanent</code>	Cert is permanent
<code>encryption_level</code>	Encryption level
<code>encryption_method</code>	Encryption method

ssl.log

FIELD	DESCRIPTION
<code>ts</code>	Time SSL first detected
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>version</code>	SSL/TLS version server chose
<code>cipher</code>	Cipher suite server chose
<code>curve</code>	ECDH/ECDSA curve if used
<code>server_name</code>	SNI hostname from client
<code>resumed</code>	Session resumed prior keys
<code>last_alert</code>	Last TLS alert seen
<code>next_protocol</code>	ALPN protocol server chose
<code>established</code>	Handshake completed successfully
<code>ssl_history</code>	Ordered handshake message codes
<code>cert_chain_fps</code>	Server cert fingerprint chain
<code>client_cert_chain_fps</code>	Client cert fingerprint chain
<code>subject</code>	Server certificate subject
<code>issuer</code>	Server certificate issuer
<code>client_subject</code>	Client certificate subject
<code>client_issuer</code>	Client certificate issuer
<code>sni_matches_cert</code>	SNI matches server cert

kerberos.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>request_type</code>	AS, TGS, or AP
<code>client</code>	Client principal
<code>service</code>	Service principal
<code>success</code>	Request succeeded
<code>error_msg</code>	Error message
<code>from</code>	Ticket valid from
<code>till</code>	Ticket valid until
<code>cipher</code>	Ticket encryption type
<code>forwardable</code>	Forwardable flag requested
<code>renewable</code>	Renewable flag requested
<code>client_cert_subject</code>	Client cert subject
<code>client_cert_fuid</code>	Client cert file UID
<code>server_cert_subject</code>	Server cert subject
<code>server_cert_fuid</code>	Server cert file UID

ftp.log

FIELD	DESCRIPTION
<code>ts</code>	Time when the command was sent
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>user</code>	FTP session username
<code>password</code>	FTP password if captured
<code>command</code>	Client FTP command
<code>arg</code>	Command argument
<code>mime_type</code>	Sniffed file MIME type
<code>file_size</code>	Transfer size if known
<code>reply_code</code>	Server reply code
<code>reply_msg</code>	Server reply message
<code>data_channel_passive</code>	Data channel uses PASV/EPSV
<code>data_channel.orig_h</code>	Data channel initiator IP
<code>data_channel.resp_h</code>	Data channel acceptor IP
<code>data_channel.resp_p</code>	Data channel acceptor port
<code>fuid</code>	File unique ID

ntp.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>version</code>	NTP version
<code>mode</code>	NTP mode
<code>stratum</code>	Stratum
<code>poll</code>	Max interval between messages
<code>precision</code>	System clock precision
<code>root_delay</code>	Delay to reference clock
<code>root_disp</code>	Dispersion to reference clock
<code>ref_id</code>	Reference clock ID
<code>ref_time</code>	Last clock set time
<code>org_time</code>	Client depart time
<code>rec_time</code>	Server receive time
<code>xmt_time</code>	Server transmit time
<code>num_exts</code>	Unparsed extension field count

smb_files.log

FIELD	DESCRIPTION
<code>ts</code>	Time file first discovered
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>action</code>	File UID
<code>fileid</code>	SMB file action
<code>path</code>	Tree path
<code>name</code>	Filename
<code>size</code>	Total file size
<code>prev_name</code>	Previous name (rename)
<code>times.modified</code>	Modification time
<code>times.accessed</code>	Access time
<code>times.created</code>	Creation time
<code>times.changed</code>	Change time

dhcp.log

FIELD	DESCRIPTION
<code>ts</code>	Time of earliest DHCP message
<code>uids</code>	Connection UIDs in the conversation
<code>client_addr</code>	Client IP
<code>server_addr</code>	Server IP
<code>mac</code>	Client hardware address
<code>host_name</code>	Hostname option (12)
<code>client_fqdn</code>	Client FQDN option (81)
<code>domain</code>	Domain name option (15)
<code>requested_addr</code>	Requested IP
<code>assigned_addr</code>	Assigned IP
<code>lease_time</code>	Lease interval
<code>client_message</code>	Client DECLINE/NAK reason
<code>server_message</code>	Server NAK reason
<code>msg_types</code>	DHCP message types in the transaction
<code>duration</code>	First-to-last message time

irc.log

FIELD	DESCRIPTION
<code>ts</code>	Time command was seen
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>nick</code>	IRC nickname
<code>user</code>	IRC username
<code>command</code>	Client IRC command
<code>value</code>	Command value/target
<code>addl</code>	Additional command data
<code>dcc_file_name</code>	DCC filename requested
<code>dcc_file_size</code>	DCC size claimed by sender
<code>dcc_mime_type</code>	Sniffed DCC MIME type
<code>fuid</code>	File unique ID

ldap_search.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>message_id</code>	LDAP message ID
<code>scope</code>	Search scope
<code>deref_aliases</code>	Alias dereference policy
<code>base_object</code>	Search base DN
<code>result_count</code>	Number of results
<code>result</code>	Result code
<code>diagnostic_message</code>	Server diagnostic
<code>filter</code>	Search filter
<code>attributes</code>	Requested attributes

postgresq.log

FIELD	DESCRIPTION
<code>ts</code>	Time of activity
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>user</code>	User from StartupMessage
<code>database</code>	Database from StartupMessage
<code>application_name</code>	App name from StartupMessage
<code>frontend</code>	Frontend/client message type
<code>frontend_arg</code>	Frontend message argument
<code>backend</code>	Backend/server message type
<code>backend_arg</code>	Backend message argument
<code>success</code>	Whether the request succeeded
<code>rows</code>	Rows returned if applicable

snmp.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>duration</code>	First-to-latest packet
<code>version</code>	SNMP version
<code>community</code>	v1/v2c community or v3 username
<code>get_requests</code>	GET varbind count
<code>get_bulk_requests</code>	GETBULK varbind count
<code>get_responses</code>	GET response varbind count
<code>set_requests</code>	SET varbind count
<code>display_string</code>	Responder sysDescr
<code>up_since</code>	Claimed boot time

radius.log

FIELD	DESCRIPTION
<code>ts</code>	Time of first packet
<code>uid</code>	Connection UID
<code>id.orig_h</code>	Originator IP
<code>id.orig_p</code>	Originator port
<code>id.resp_h</code>	Responder IP
<code>id.resp_p</code>	Responder port
<code>username</code>	Username
<code>mac</code>	MAC address
<code>framed_addr</code>	Assigned address hint
<code>tunnel_client</code>	Tunnel-Client-Endpoint
<code>connect_info</code>	Connect-Info
<code>reply_msg</code>	Server challenge/reply text
<code>result</code>	Success or failure
<code>t1l</code>	Time from first request to Accept/error

ldap.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
message_id	LDAP message ID
version	LDAP version
opcode	LDAP operation
result	Result code
diagnostic_message	Server diagnostic
object	Target object
argument	Operation argument